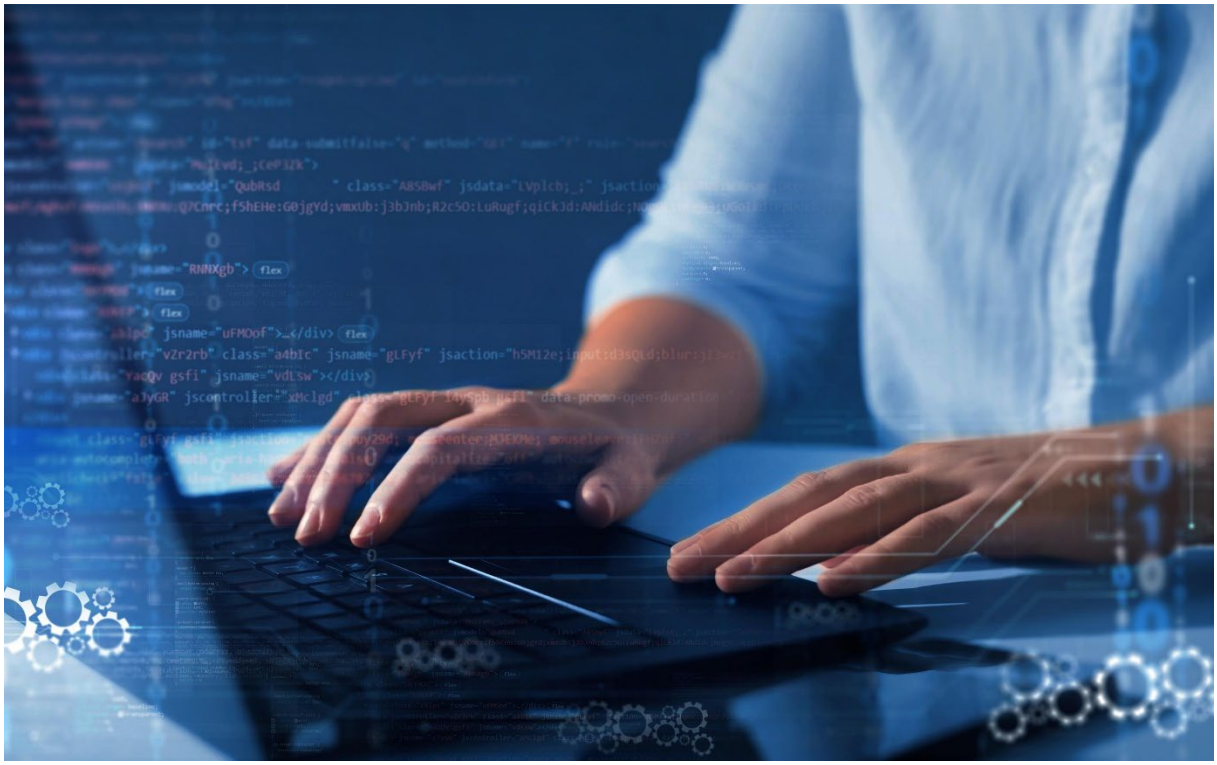
GELOS
ENTERPRISES



Coding and Testing Guidelines

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1. About these guidelines

1.1 Document purpose

These guidelines serve as a reference for developers within Gelos Enterprises (Gelos) to ensure consistency, readability and maintainability of code developed in projects. It outlines best practices, naming conventions and documentation standards that all development teams should follow.

1.2 Scope

This document applies to all employees of Gelos.

2. Coding guidelines

This section outlines the guidelines to follow when developing code.

2.1 Best practices

Apply the following best practices in your projects:

- Coding should be planned first by designing the algorithm in pseudocode, a flowchart or a similar design tool and included in planning documentation, where relevant.
- Add comments to explain complex code; avoid over-commenting obvious code.
- Follow consistent naming conventions and use meaningful names for coding elements.
- Include error handling and provide meaningful error messages.
- Ensure code is maintainable and efficient by being clear and concise.

2.2 Coding standards

Refer to and apply the following standards, relevant to the language used in your project. These include naming conventions and code layout:

- C#: C# Coding Standards and Best Practices (dofactory)
- JavaScript: JavaScript Standard Style (standard JS)
- Python: PEP 8 – Style Guide for Python Code (Python Software Foundation)
- HTML: Guidelines for writing HTML code examples (MDN Web Docs)
- PHP: PHP Rules (PHP).

2.3 Internal documentation standards

Refer to and apply the following standards for internal documentation, relevant to the language used in your project:

- C#: XML comments Annex D Documentation comments
- JavaScript: JSDoc comments @use JSDoc.

3. Testing code

This section outlines the guidelines to following when testing code.

3.1 Designing tests

Apply the following guidelines where possible or relevant when designing tests for code.

3.1.1 Test coverage

Ensure comprehensive test coverage by including unit, integration and acceptance tests. This helps in verifying that all parts of the code work as expected.

3.1.2 Test cases

Write clear and concise test cases that cover both typical and edge cases. Each test case should have a clear objective and expected outcome.

3.1.3 Automation

Automate tests wherever possible to facilitate continuous integration (CI) and continuous deployment (CD). This helps in catching issues early in the development cycle.

3.1.4 Performance testing

Include performance tests to ensure the application meets performance requirements under expected load conditions.

3.2 Documenting tests

Apply the following guidelines where possible or relevant when documenting tests for code. Use the Gelos Software Test (docx) template to document tests, where possible.

3.2.1 Test plan

Create a detailed test plan that outlines the scope, objectives, resources and deliverables of the testing activities.

3.2.2 Test cases documentation

Document each test case with a description, preconditions, steps to execute and expected results. This helps in understanding the purpose and execution of each test.

3.2.3 Test results

Maintain a record of test results, including pass or fail status, defects found and any deviations from expected outcomes. This aids in tracking the progress and effectiveness of testing.

3.2.4 Test code comments

If automated tests are used, add comments in the test code to explain the purpose and logic of complex test cases. This makes it easier for other developers to understand and maintain the tests.

4. Document information and review

4.1 Responsibilities

The following table provides a list of the positions or departments that are responsible for this policy and procedure and the scope of their responsibility.

Table 1 – Responsibilities

Position	Responsibility
CEO	Approver for these guidelines.
ICT Services Senior Manager	Maintenance of these guidelines.

4.2 Document review

This policy and procedure will be reviewed every 12 months.

Review date: 15/05/20XX

4.3 Document authority

This Coding and Testing Guidelines has been authorised by Catherine Dunn the CEO of Gelos Enterprises and is available to all staff. It has been developed in line with all relevant legislation, in consultation with committee representatives and will be revised on a regular basis.

Approval date: 15/05/20XX